

Installation Guide

Node-RED Flow Configuration and Deployment Manual

Installation and Setup

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1. Raspberry Pi OS Setup

Raspberry Pi OS (32-bit/64-bit, Debian Bookworm)

Flash Raspberry Pi OS to the SD card, enable SSH, and optionally set a static IP.

Raspberry Pi Initial Setup

Begin by ensuring your Raspberry Pi operating system is up to date.

Update System Packages:

```
sudo apt update && sudo apt upgrade -y
```

Network Configuration

Configure a static IP address to ensure consistent connectivity to the servers and for remote access to the dashboard.

Configure Static IP:

1. Right-click on the Wi-Fi icon in the taskbar
2. Go to Advanced Options → Edit Connections
3. Select Ethernet → Wired Connection
4. Click the Add button
5. Enter the desired IP Address and Netmask
6. Ensure the IP can ping your server
7. Click Save
8. Reboot the system for changes to take effect

2. Node-RED Installation

Node-RED is installed using the official installation script which also installs the correct version of Node.js.

2.1. Install Node-RED:

```
bash <(curl -sL https://raw.githubusercontent.com/node-red/linux-installers/master/deb/update-nodejs-and-nodered)
```

If the installer fails to install Node.js, manually install it:

```
curl -sL https://deb.nodesource.com/setup_20.x | sudo -E bash - sudo apt install -y nodejs
```

Verify Installation:

```
node -v  
npm -v
```

If Node.js installation fails:

```
curl -sL https://deb.nodesource.com/setup_20.x | sudo -E bash -&sudo apt install -y nodejs
```

2.2. Start Node-RED:

For manual start:

```
node-red-start
```

To enable auto-start on boot:

```
sudo systemctl enable nodered.service
```

To disable auto-start:

```
sudo systemctl disable nodered
```

Check Service Status:

```
sudo systemctl status nodered.service
```

2.3. Access Node-RED Web Interface:

Open a web browser and navigate to the URL where your Node-RED Dashboard is available. By default, it is often at

<http://localhost:1880/ui>

3. MQTT Broker Configuration

Mosquitto MQTT broker provides the messaging backbone for the system, enabling communication python bridge scripts and Node-RED flows.

Install Mosquitto:

```
sudo apt update
sudo apt install mosquitto mosquitto-clients
```

Enable and Start Service:

```
sudo systemctl enable mosquitto
sudo systemctl start mosquito
```

3.1. Configure MQTT Bridge:

To bridge with a remote MQTT broker (for cloud connectivity), create a bridge configuration:

```
sudo nano /etc/mosquitto/conf.d/bridge.conf
```

Add the following configuration (modify address and port as needed):

```
connection bridge-remote
address 127.0.0.1:1883
topic # both 0
```

Restart Mosquitto:

```
sudo systemctl restart mosquito
```

3.2. Verify Bridge Configuration:

Test the bridge with two terminal windows:

Terminal 1 (Subscribe):

```
mosquitto_sub -h localhost -t my/topic
```

Terminal 2 (Publish):

```
mosquitto_pub -h 127.0.0.1 -p 1883 -t my/topic -m "Hello from remote broker"
```

If the bridge is working correctly, the message will appear in Terminal 1.

4. Python Virtual Environment Setup

The Python-based bridge script connects to SINUMERIK/FANUC servers on industrial equipment, reads process variables, and publishes them to the local MQTT broker for consumption by Node-RED.

4.1. Create Virtual Environment:

Use a virtual environment to isolate project dependencies:

```
python3 -m venv myenv
```

4.2. Activate Virtual Environment:

```
source myenv/bin/activate
```

Install Libraries in Virtual Environment:

```
pip install opcua paho-mqtt numpy
```

4.3. Deactivate Virtual Environment:

When finished working in the virtual environment:

```
Deactivate
```

5. Kiosk Mode & Auto start

Configure the system to automatically start the python bridge script and display the dashboard in kiosk mode on boot.

Step 1: Create Kiosk Startup Script

Create the file:

```
nano /home/cmti/myenv/lib/python3.11/site-packages/start_kiosk.sh
```

Add the following content:

```
#!/bin/bash
Sleep 25
chromium-browser --start-fullscreen --kiosk http://localhost:1880/ui
```

Make the script executable:

```
chmod +x /home/cmti/myenv/lib/python3.11/site-packages/start_kiosk.sh
```

Step 2: Save Bridge Script

Save your python bridge script (e.g., `opcua_mqtt_bridge.py`) in the same directory:

```
/home/cmti/myenv/lib/python3.11/site-packages/opcua_mqtt_bridge.py
```

Step 3: Create Autostart Configuration

Navigate to home directory and show hidden files:

1. Right-click in the home directory
2. Select "Show Hidden"
3. Navigate to the `.config` folder
4. Create a new folder named "autostart" if it doesn't exist
5. Create a file named "autorun.desktop" in the autostart folder

Step 4: Configure Autostart Entries

Add two entries to `autorun.desktop`:

Entry 1: Kiosk_Mode.sh

```
[Desktop Entry]
Type=Application
Name=Autorun Kiosk
Exec=/home/pi/my_env/lib/python3.11/site-packages/start_kiosk.sh
StartupNotify=false
Terminal=false
Hidden=false
```

Entry 2: OPCUABridge.sh

```
[Desktop Entry]
Type=Application
Name=GUI Controller
Exec=python3 /home/pi/my_env/lib/python3.11/site-packages/opcua_mqtt_bridge.py
StartupNotify=false
Terminal=true
Hidden=false
```

Step 5: Reboot System

Reboot the Raspberry Pi to activate kiosk mode:

```
sudo reboot
```

6. Node-RED Flow Import

Follow these steps to import Node-RED flow configurations:

1. Open web browser and navigate to `http://localhost:1880`
2. Click the menu (three horizontal bars) in the top-right corner
3. Select "Import" from the menu
4. In the "Import nodes" dialog, paste the JSON flow code
5. Click "Import" to add the flow to the workspace
6. Install any missing nodes from the Palette Manager
7. Configure node parameters (double-click nodes to edit)
8. Click "Deploy" in the top-right corner

7. Troubleshooting

Node-RED not starting	• Check Node-RED service status: <code>sudo systemctl status nodered.service</code> • View logs: <code>journalctl -u nodered -f</code> • Ensure Node.js is installed correctly: <code>node -v</code> • Try manual start: <code>node-red-start</code>
MQTT Bridge not working	• Verify Mosquitto is running: <code>sudo systemctl status mosquitto</code> • Check broker configuration: <code>/etc/mosquitto/conf.d/bridge.conf</code> • Test local connection: <code>mosquitto_sub -h localhost -t test/topic</code> • Check firewall rules if connecting to remote broker
Server connection fails	• Verify server is reachable: <code>ping <server-ip></code> • Check username and password credentials • Verify network configuration (static IP) • Check Python script for correct server address • Review OPC UA server security policies
Dashboard not accessible	• Verify Node-RED is running on port 1880 • Check firewall rules: <code>sudo ufw status</code> • Ensure flows are deployed • Clear browser cache and retry • Check Node-RED logs for errors
Dashboard not accessible	• Verify <code>autorun.desktop</code> file exists in <code>~/.config/autostart/</code> • Check file permissions: <code>chmod +x start_kiosk.sh</code> • Verify paths in desktop entries are correct • Check system logs: <code>journalctl -xe</code>
